Experiment 3

Flow and Thermal Behaviour in a Centrifugal Pump Passage

Aim

To evaluate centrifugal pressure development and temperature variation in a simplified rotating annular passage at several pump speeds.

Simulation Parameters

Parameter	Value
Coordinate system	Cylindrical
Radial grid points	80
Azimuthal grid points	160
Reynolds number	500
Iterations	1200
Pump speeds	1000, 2000, 3000 rpm

Numerical Method

A reduced-order model in a rotating polar frame prescribes radial and tangential velocity components. The model includes centrifugal pressure rise, wall heat exchange, and heating associated with losses.

Boundary Conditions

The inlet has a specified temperature and radial inflow at the pump eye. Flow moves outward radially, the azimuthal direction is periodic, and wall heat transfer is represented by a distributed source term.

Observations

Pressure contours, estimated pump head, absolute and relative speed, inlet/outlet temperatures, and thermal residual histories were compared across the three rotational speeds.

Results

Speed (rpm)	Re	Inlet temp. (K)	Outlet temp. (K)	Thermal residual	Peak pressure scale
1000	500	300	300.30	3.56×10	7×10^{6}
2000	500	300	300.18	1.81×10	2.8×10^{6}
3000	500	300	300.12	1.22×10	2.8×10^{6}
5151	500	300	300.086	7.1258×10^{-5}	1.9581×10^{6}

Conclusion

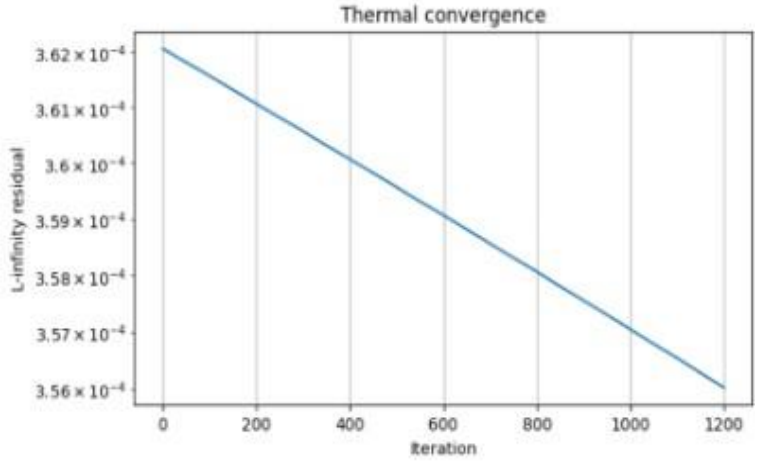
The pressure field strengthens as rotational speed increases, consistent with greater centrifugal pressure generation. The reported maximum pressure scale rises from about 7×10 at 1000 rpm to about 2.8×10 at 2000 and 3000 rpm. The predicted outlet-temperature increase remains modest, and the thermal residual decreases across the tested speeds.

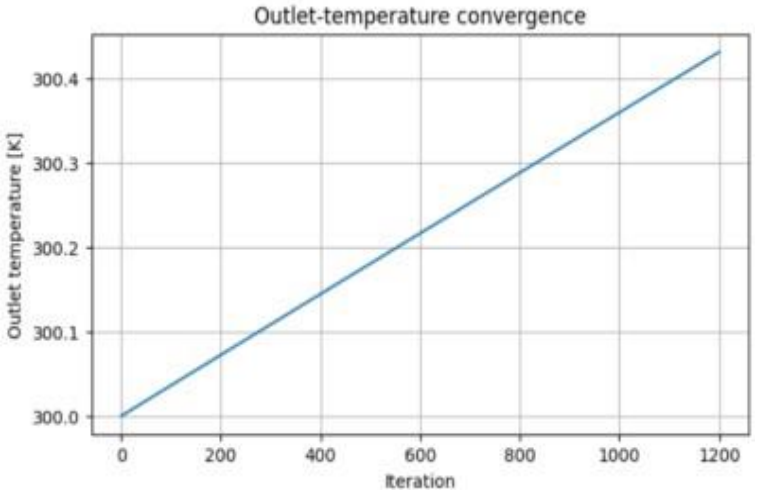
Original Simulation Figures

The following pages contain the plots and simulation outputs from the supplied report.

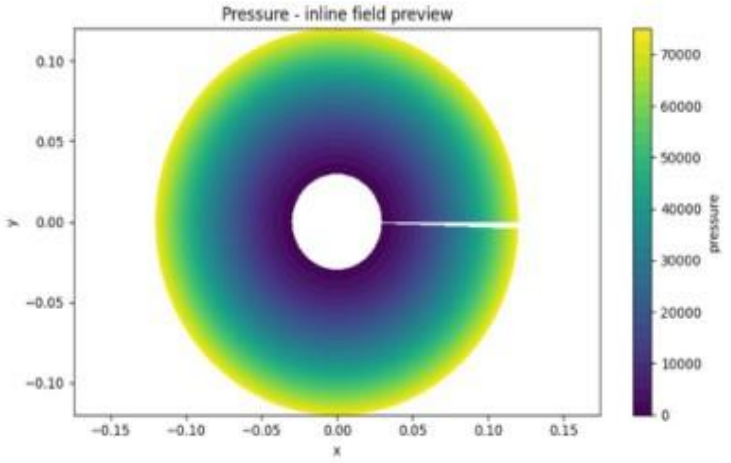
Original simulation output (source page 2)

For pump rotation at 1000 rpm:

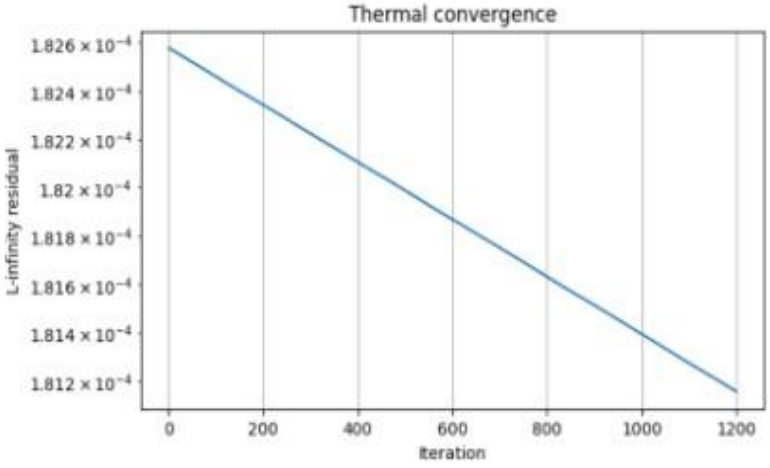




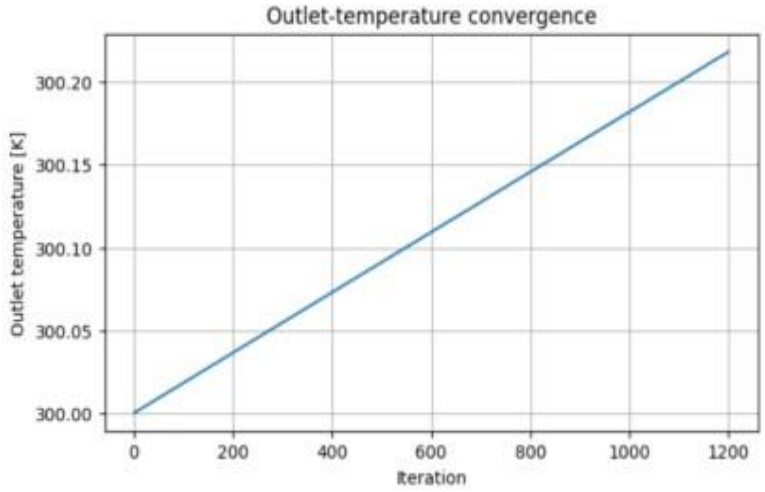
Original simulation output (source page 3)

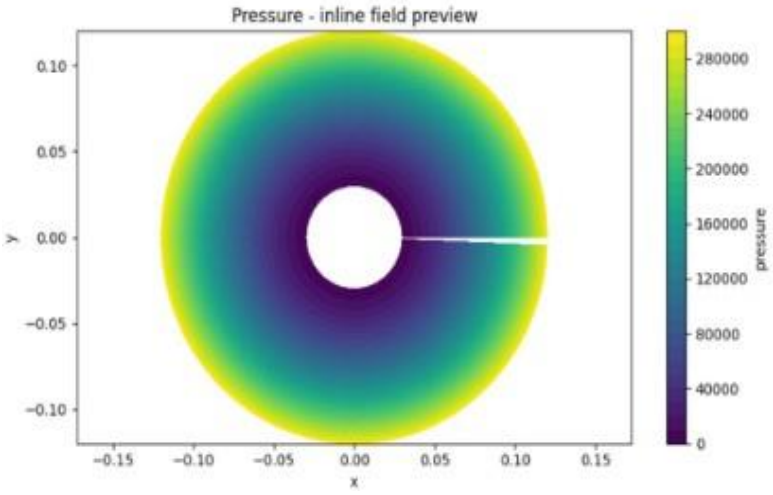


For pump rotation at 2000 rpm:



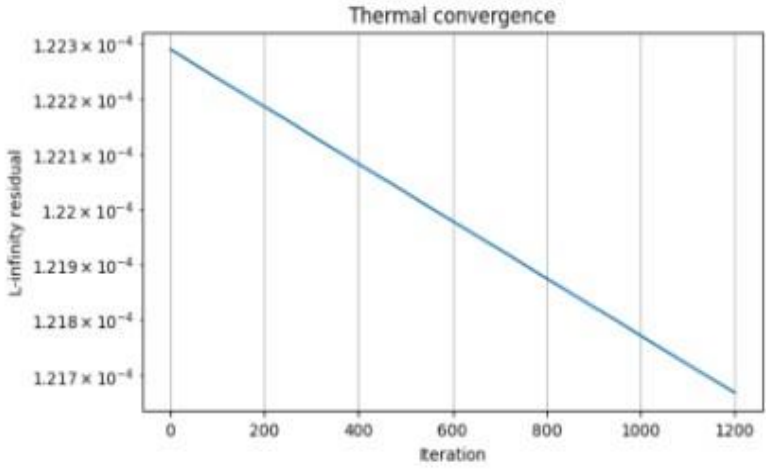
Original simulation output (source page 4)

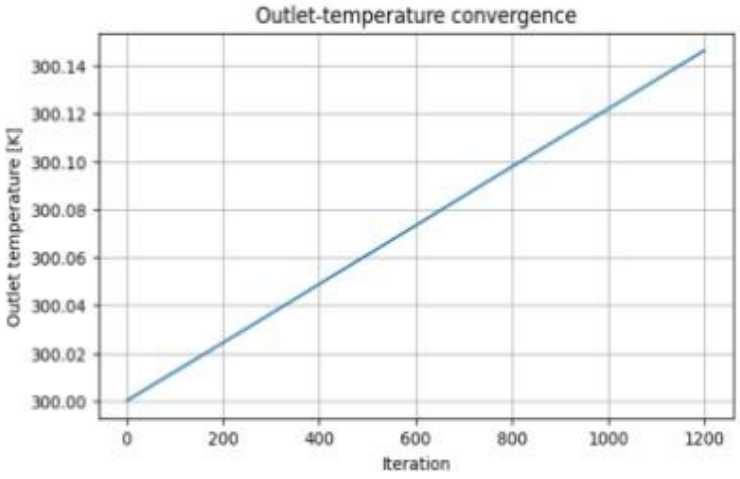




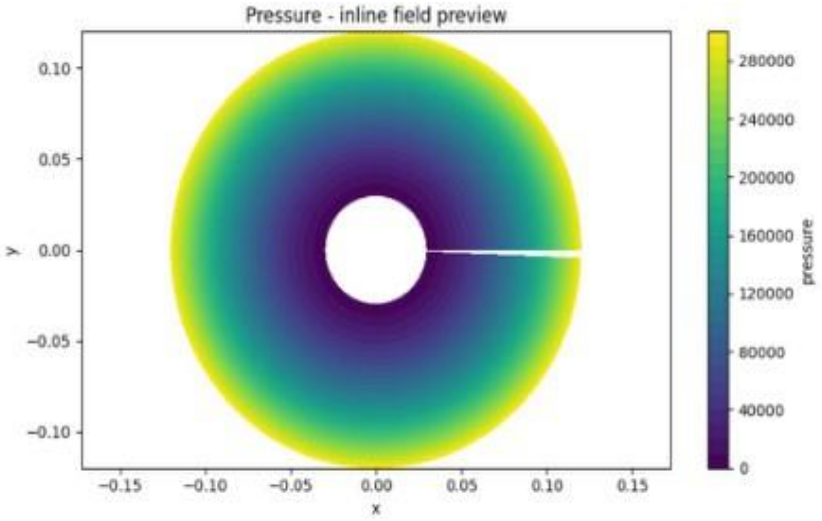
Original simulation output (source page 5)

For pump rotation at 3000 rpm:

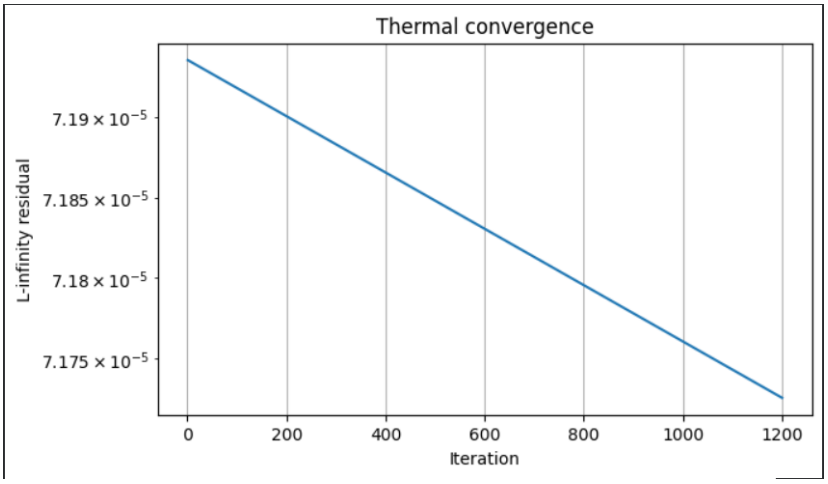


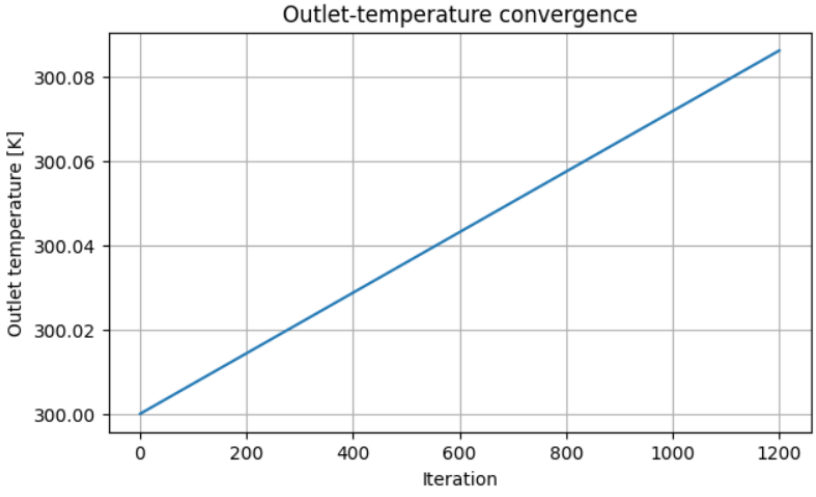


Original simulation output (source page 6)

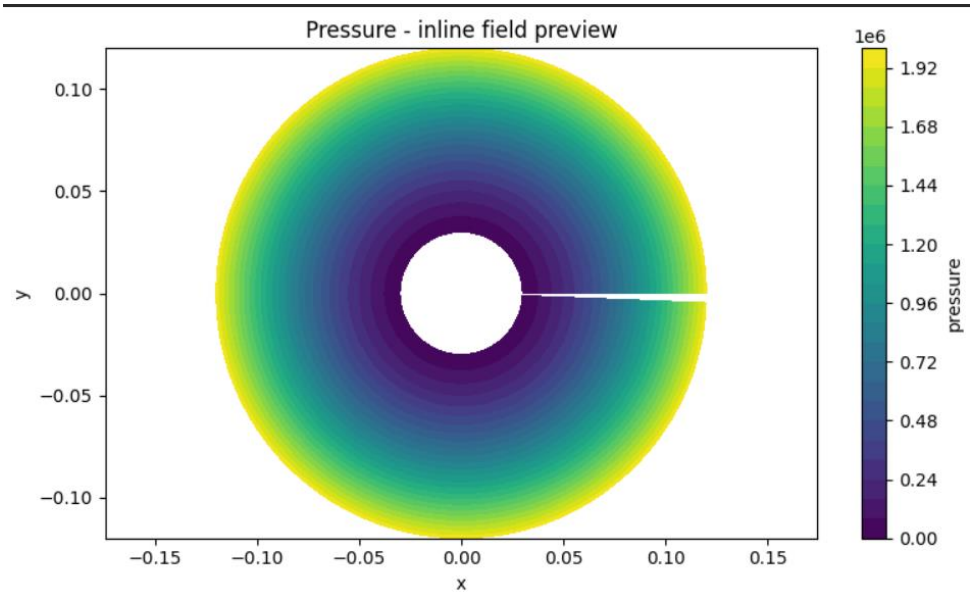


For pump rotation at 5151 rpm:





Original simulation output (source page 7)



Results and Discussion

A simplified two-dimensional annular single-rotating-frame centrifugal pump heat-transfer model was analysed numerically at a fixed Reynolds number of 500. The inlet temperature was maintained at 300 K, while the rotational speed was varied to study its effect on the thermal response and pressure rise of the pump.

At 1000 RPM, the outlet temperature was 300.30 K, with a reported thermal residual of 3.56×10^{-n} and a peak pressure scale of 7×10^{n}. At 2000 RPM, the outlet temperature decreased to 300.18 K, while the thermal residual was 1.81×10^{-n} and the peak pressure scale was 2.8×10^{n}. At 3000 RPM, the outlet temperature was 300.12 K, with a thermal residual of 1.22×10^{n} and a peak pressure scale of 2.8×10^{n}. The exponents in these reported residual and pressure values should be verified from the original solver output.

An additional simulation was performed at 5151 RPM. The solver reported an outlet temperature of 300.0862 K, a pressure rise of 1.9581×10^{6} Pa, a pump head of approximately 200.02 m, and a final L-infinity residual of 7.1258×10^{-5}.

For the 5151 RPM case, the reported fluid heat gain was approximately 16.98 W, and the mass flow rate was approximately 0.0471 kg/s. These results describe the model response at the selected operating condition.

The additional high-speed result provides a further data point for examining the relationship between rotational speed and pump performance. However, the expected near-linear relationship between pump head and RPM² should be evaluated using the pump-head values obtained from all speed cases.

Conclusion

The centrifugal pump heat-transfer model was investigated at a fixed Reynolds number of 500 and an inlet temperature of 300 K for different rotational speeds. The reported outlet temperatures for 1000, 2000, and 3000 RPM were 300.30 K, 300.18 K, and 300.12 K, respectively. For the additional case at 5151 RPM, the solver predicted an outlet temperature of 300.0862 K.

At 5151 RPM, the numerical model reported a pressure rise of approximately 1.9581 MPa, a pump head of approximately 200.02 m, and a final L-infinity residual of 7.1258×10^{-5}. The reported fluid heat gain was approximately 16.98 W.

Original simulation output (source page 8)

The results demonstrate how the simplified numerical model can be used to investigate the thermal and pressure-related performance of a centrifugal pump at different rotational speeds. A quantitative assessment of pump-head scaling with RPM^2 requires consistent pump-head results for all operating speeds. The reported values should also be checked against the original solver output before final submission, particularly the thermal residuals and peak-pressure values for the first three cases.